

But what was Berkeley RISC?

In the past, CISC microprocessor cores like Motorola 68000 used only a subset of the available instruction set; therefore most of the decoding circuitry for the never-used instructions was wasted. But in RISC, CISC was replaced with multiple registers to speed up the execution since access to these registers would take less cycles than memory access needs, outperforming CISC. Considering the case of ADD instructions in which CISC would appear in different formats where one would be meant for adding the numbers in two registers and placing the result in third, whilst the other would allow to add two numbers stored in the main memory and store the result in a register, offering wide variety of instructions to perform the required operations and providing several options as operands.

But in the case of RISC designs, the ADD would always take registers as operands, which creates a necessity for the programmer to write additional instructions to load the values from memory whenever required. Now, the Berkeley RISC was able to provide good performance utilising pipelining and register-windowing technique. In register windowing, the CPU contains a large number of registers around 128, together called a register file, whilst the group of eight registers is called a window. But a program can only use one window, thereby allowing fast procedure calls. The call simply moves the window down to the set of eight registers used by that procedure and the return moves the window back.

After that time, RISC has seen a lot of developments proving its simplicity over CISC. Here are some of the features of RISC:

1. Large general-purpose 32-bit register banks
2. Fixed 32-bit instruction size
3. Hard-wired instruction decode logic instead of microcoded ROMs
4. Single-cycle execution is possible
5. Pipelined execution
6. Easier to prototype

Processor Mode and Mode Number

Processor mode	Mode number	Description
User (usr)	0b10000	Normal program execution mode
FIQ (fiq)	0b10001	Supports a high-speed data transfer or channel process
IRQ (irq)	0b10010	Used for general-purpose interrupt handling
Supervisor (svc)	0b10011	A protected mode for the operating system
Abort (abt)	0b10111	Implements virtual memory and/or memory protection
Undefined (und)	0b11011	Supports emulation of hardware coprocessor
System (sys)	0b11111	Runs privileged operating system tasks (ARMv4 and above)

RISC features can be introduced in CISC processors but would require much more hardware. A typical RISC architecture consists of a large uniform register file, load and store architecture, simple addressing modes and uniform fixed-length instruction fields. Due to this characteristic, we achieve high performance, low code size, low power consumption and low silicon area.

What features does ARM provide?

British multinational semiconductor and software design company, ARM Holdings plc, offers complete solutions that are essential for the manufacturing process, but the company does not manufacture ICs. Moreover, it provides software development tools under the brand Keil and RealView. It also provides ARM architecture licensing for the companies that want to manufacture ARM-based CPUs or 'system-on-chip' products.

The ARM architecture is a simple hardware design allowing things to be left off the chip. It helps in creating small die-sized chip which helps in considerably reducing the cost. Its low cost, simple pipeline construction and the freedom to put the design point where the designer finds it suitable for low-power consumption adds onto the benefits it provides for embedded applications.

ARM allows an instruction set called 'Thumb', which compresses 32-bit instructions to 16 bits, enabling programs to be coded much more densely than standard RISC instruction sets. Processors enabled to execute 'Thumb' also allow 32-bit instructions to run on the same hence allowing 16-bit and 32-

bit instructions to mix together without affecting the performance, maintaining powerful computing capabilities.

ARM designs and licences intellectual property (IP) instead of manufacturing the chips. Two types of licences that ARM provides are 'implementation licence' and 'architecture licence.' Implementation licence provides information to design integrated circuit containing the ARM core. The architecture licence enables the licencees to design their own processor with ARM instruction set architecture.

ARM cores are simple compared to other processors since it can be manufactured with few transistors. ARM is a 32-bit instruction set architecture. Today, many embedded applications like smartphones, set-top boxes, digital televisions and digital cameras use ARM processors due to their cost-effectiveness and low-power consumption. ARM architecture is compatible with all four major operating systems, i.e., Symbian OS, Palm OS, Windows and Android OS.

The instruction sets in ARM processor are classified as ARM instruction set, Thumb instruction set and Jazelle mode. ARM mode is a standard 32-bit instruction set. Thumb instruction set is a 16-bit compressed form that provides better performance than complex instruction set computers (CISCs). Jazelle DBX (direct bytecode execution) allows some ARM processors to execute Java bytecode.

An ARM processor consists of 31 general-purpose 32-bit registers. Sixteen registers, namely R0-R15, are visible, which means they can be modified by the user whereas other registers help to speed up the execution process.